Explainable Machine Learning for Water Distribution Systems

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About me



- Post-doc in Machine Learning Group @ Bielefeld University, Germany
  - PhD in Computer Science (focus on AI + ML)
- Visiting Researcher @ University of Cyprus

Research focus:

- Foundations Explainable AI (XAI) => Counterfactual Explanations
- Trustworthy AI in critical systems
- X(AI) in Water Distribution Systems

Outline

- Potential of Machine Learning (ML) in Water Distribution Systems (WDS)
- Trustworthy AI and Explainable ML
  - Introduction to Explainable AI (XAI)
  - Case-1: Interpretable Event Diagnosis
  - Case-2: AI-based Surrogates for efficient What-If Analysis

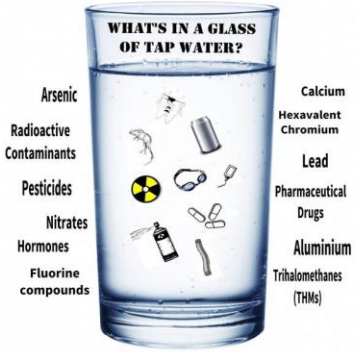
- Hands-on in Python
  - Case-1 and 2 in Python

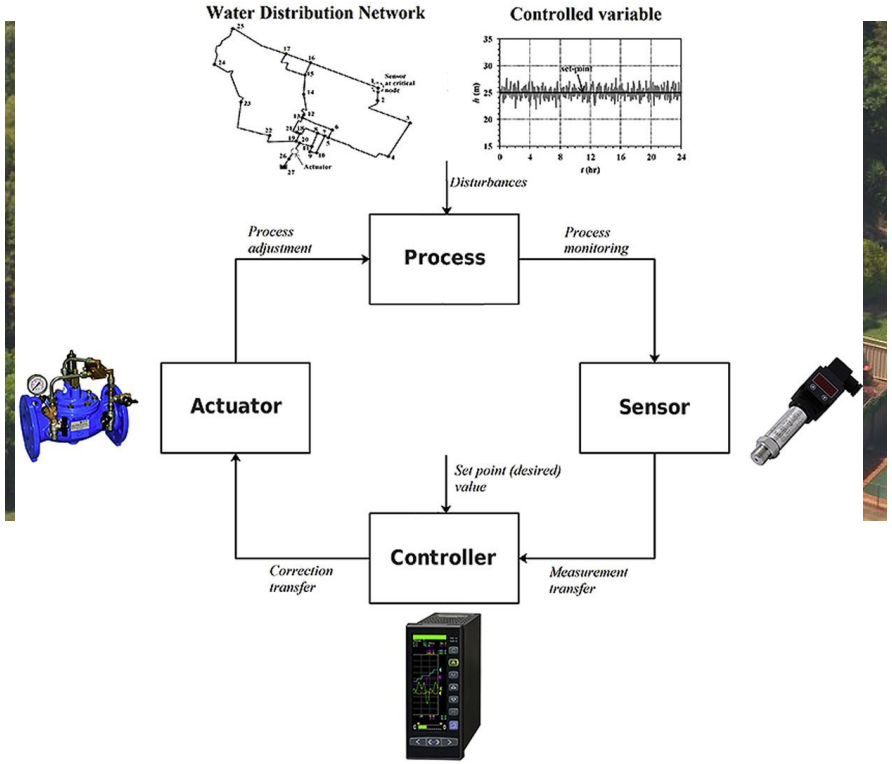
Potential of ML in WDS

Tasks of Water Utilities

- Satisfy water demands at all times
- Ensure water quality – safe to drink
- Network planning
- Demand forecasting
- Leakage detection & localization
- Contamination detection & localization
- Control
- ….



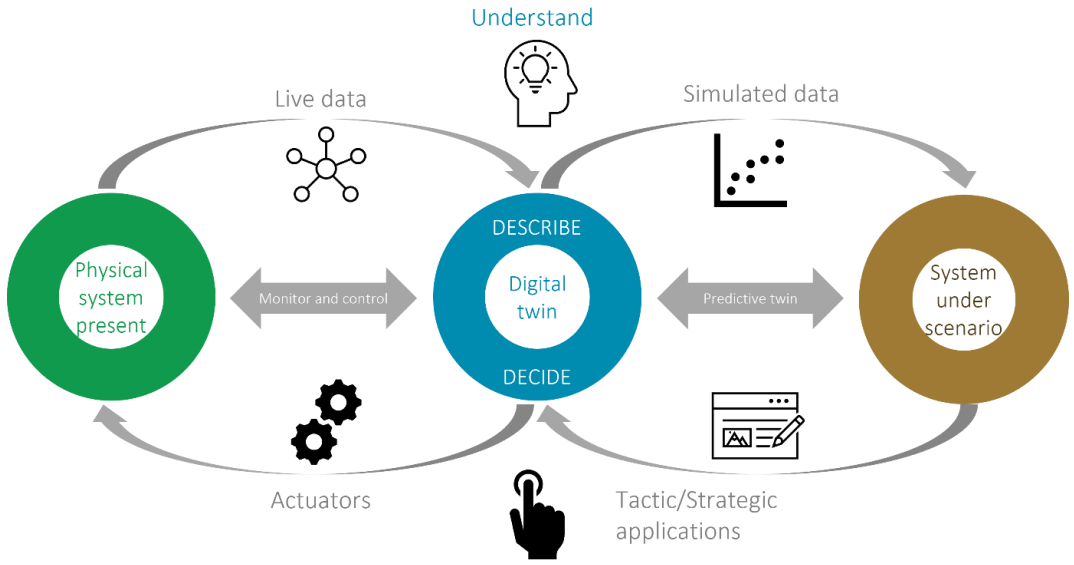




Applications of AI/ML in WDNs

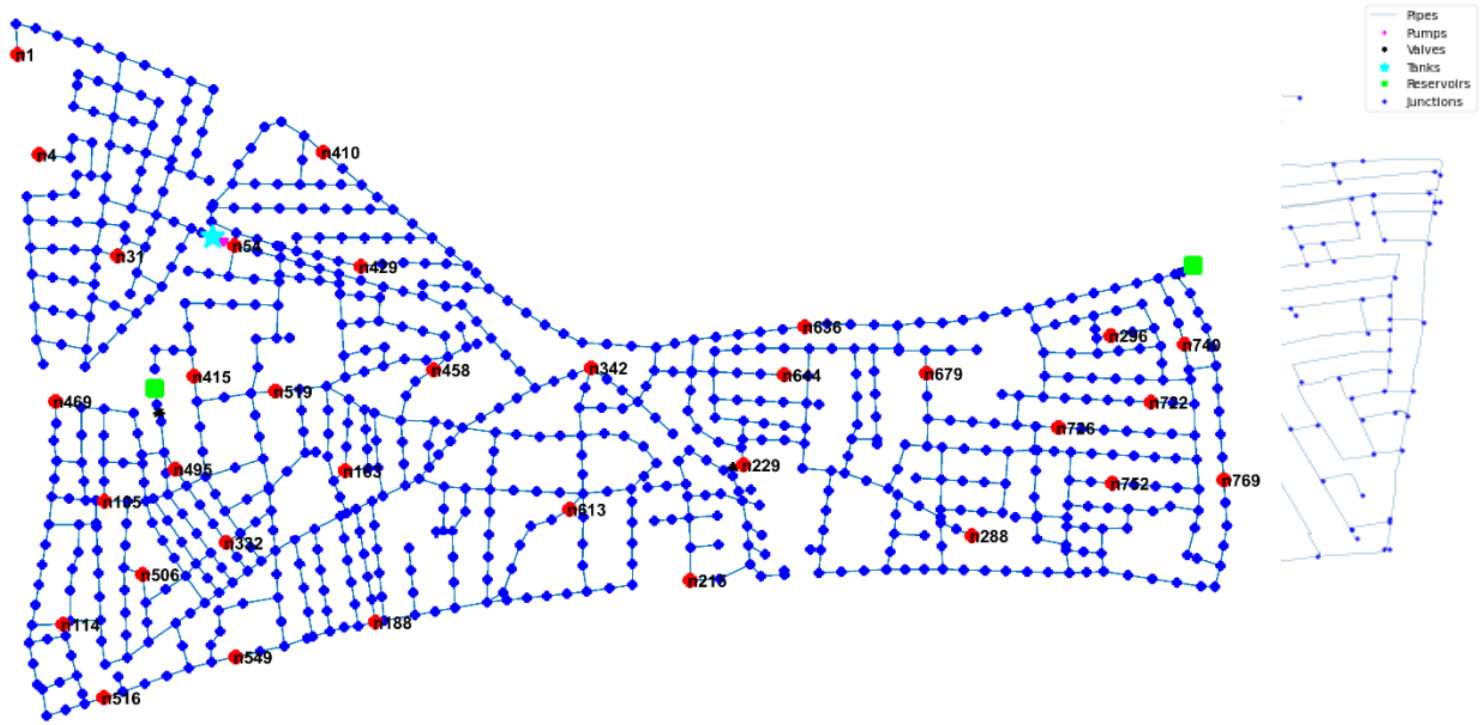
Operation of WDNs is non-trivial!
e.g. different events, complex dynamics, etc.
=> **How can AI help?**

- Event detection and isolation
- Forecasting
- Control
- Surrogates & Emulators for complex simulations
  => e.g. Digital twins



Challenges

- Lots of uncertainties
  - Topology
  - Measurements
- Sparse sensor readings
- Long time horizons
  - Planning/Forecasting
  - Policies
  - Transport delay
- High-stakes application
  - Trustworthiness + Liability



Trustworthiness and Explainability

- **What is needed for a successful and responsible deployment of AI/ML?**

=> *Trustworthiness*

- Understanding as foundation of trust
  - Many dimensions: *Verifiability, robustness, fairness, traceability, interpretability, ...*
- **"Plain" AI/ML system are not trustworthy**
- Should not be applied out of the box
- *"Extensions" are needed!*

=> Explainable AI/ML for investigating and enabling trustworthiness

Introduction to XAI

Very brief!

Need for Explanations

ChatGPT OpenAI



https://youtu.be/eMx-2s7mZ24



AI is out there…being deployed…busily making mistakes !

The Many Dimensions of XAI

Different needs and different goals

• Goal: Improve the system • Who: Engineer	• Goal: Build trust in the AI • Who: Lay users
• Goal: Justification • Who: Regulation authorities	• Goal: Knowledge-mining • Who: Domain experts



=> No general silver bullet! Pick the right tool for the right application!

Overview of XAI Methods

Ante-Hoc

concepts

prototypes

proxies

twin-systems

Post-Hoc

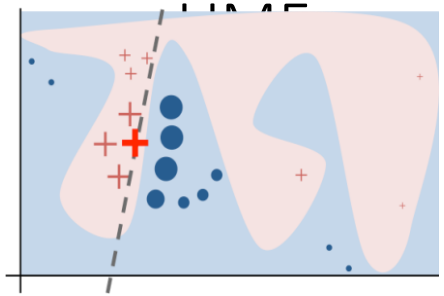
NLP

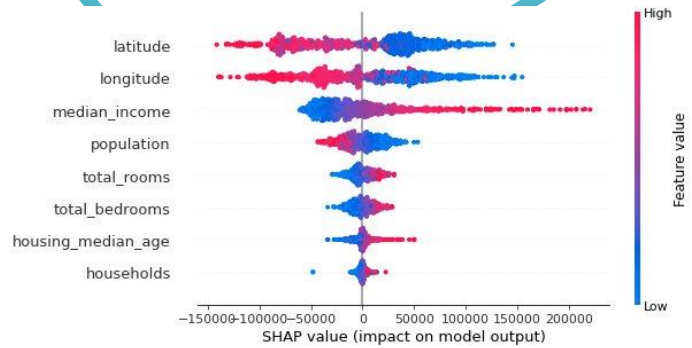
dialogue

Visualisations

feature-importance

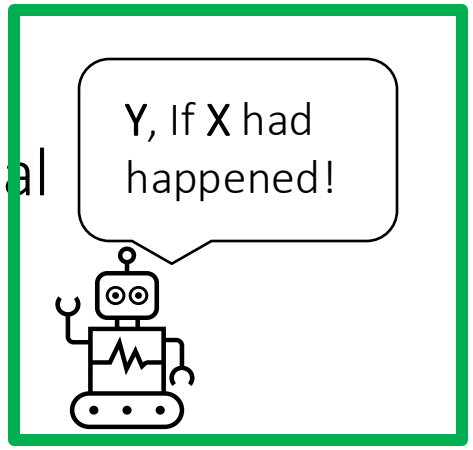
example-based





i-factual

counterfactual

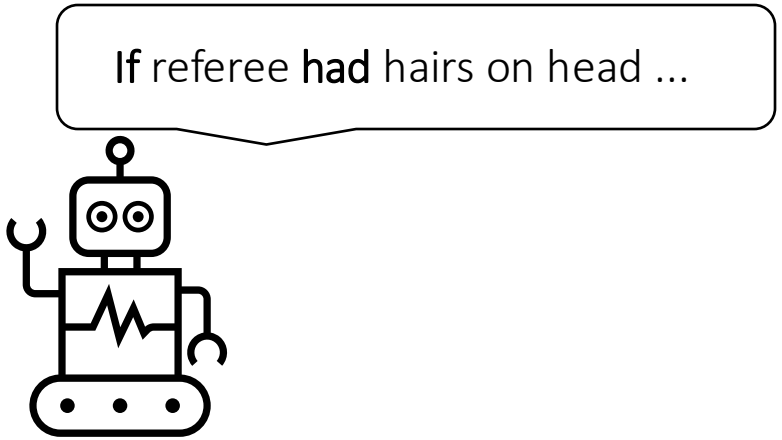


Counterfactual Explanations

- *"What-If"* explanations
- *Contrasting* explanations
  - How to change the outcome?
- *Intuitive* to humans
  - Well-grounded in philosophy, psychology, cognitive science



https://youtu.be/eMx-2s7mZ24



Formalization of Counterfactuals

Basic desiderata:

- **Contrasting property:** Outcome must change!
- **Cost/Complexity:** Prefer easy and simple explanations

Optimization problem [Wachter 2017]

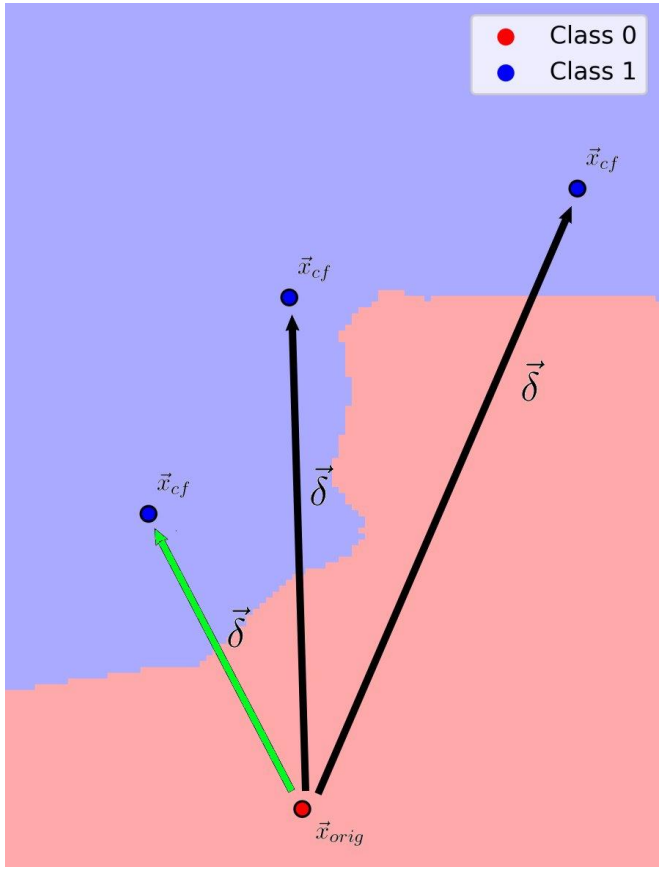
$$\underset{\vec{x}_{\text{cf}} \in \mathbb{R}^d}{\arg\min} \; \ell\left(h(\vec{x}_{\text{cf}}), y_{\text{cf}}\right) + C \cdot \theta(\vec{x}_{\text{cf}}, \vec{x}_{\text{orig}})$$

Contrastive — Cost/Complexity

$$\text{Change } \vec{\delta} = \vec{x}_{\text{cf}} - \vec{x}_{\text{orig}}$$

$$\underset{\vec{x}_{\text{cf}} \in \mathbb{R}^d}{\arg\min} \; \theta(\vec{x}_{\text{cf}}, \vec{x}_{\text{orig}}) \quad \text{s.t.} \quad h(\vec{x}_{\text{cf}}) = y_{\text{cf}}$$

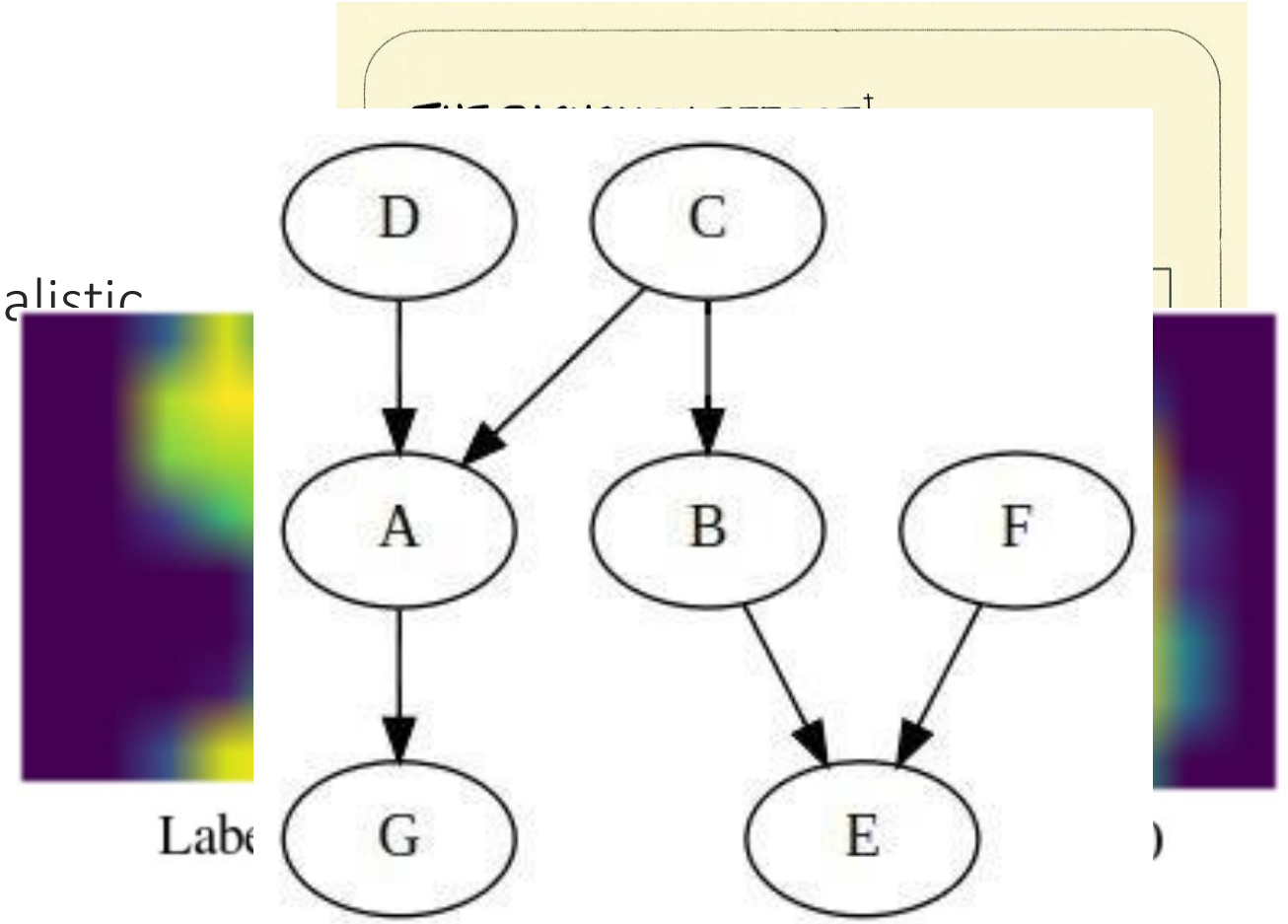
Cost/Complexity — Contrastive



Other Aspects and Desiderata

- Counterfactual are not unique
  - *Rashomon effect*
    - => Which one should we pick?
- Counterfactuals must be plausible & realistic
    - => Causality?
- Robustness
  - Input perturbations
  - Model changes
  - Noisy executions

    => *How to handle uncertainties?*
- ….

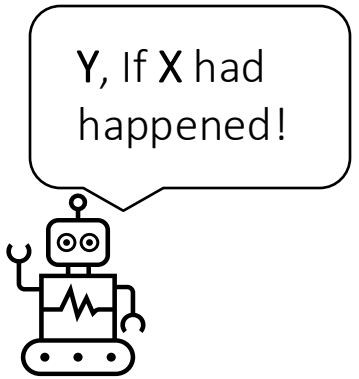


XAI in Water Distribution Systems

- Investigate importance and sensitivity of input variables
  - *Which sensors (locations) are most relevant?*
  - *Which consumer demands have the largest impact?*
- Explain predictions and outcomes
  - *Why was an alarm raised?*
  - *Why a leakage and not a sensor fault?*
- **Thinking bigger:** XAI as part of a toolchain for analysis and planning in multi-component systems
  - Event localization: *Where is the leakage?*
  - Control policies: *Chlorination control or scheduling*

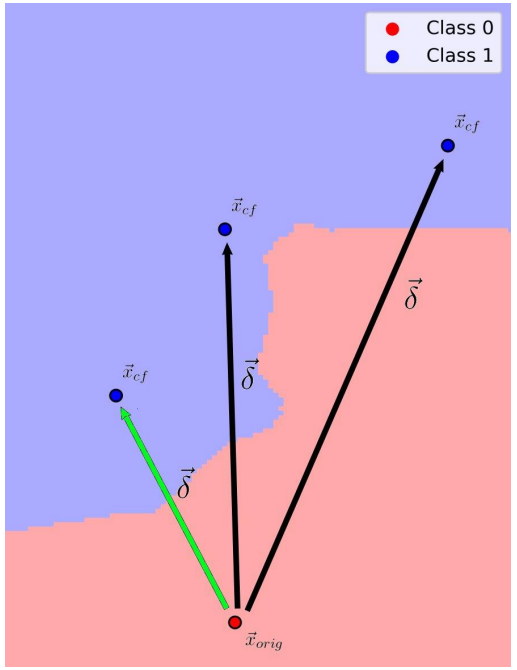
XAI -- Summary

- Many different XAI methods exists
  - *Pick the right methods for the right application!*
- Counterfactual explanations:
  - Contrasting – how to change the outcome
  - Intuitive to humans
  - Many additional aspects:
    - *Plausibility & Actionability*
    - *Robustness*
    - *…..*



=> In-depth tutorials:

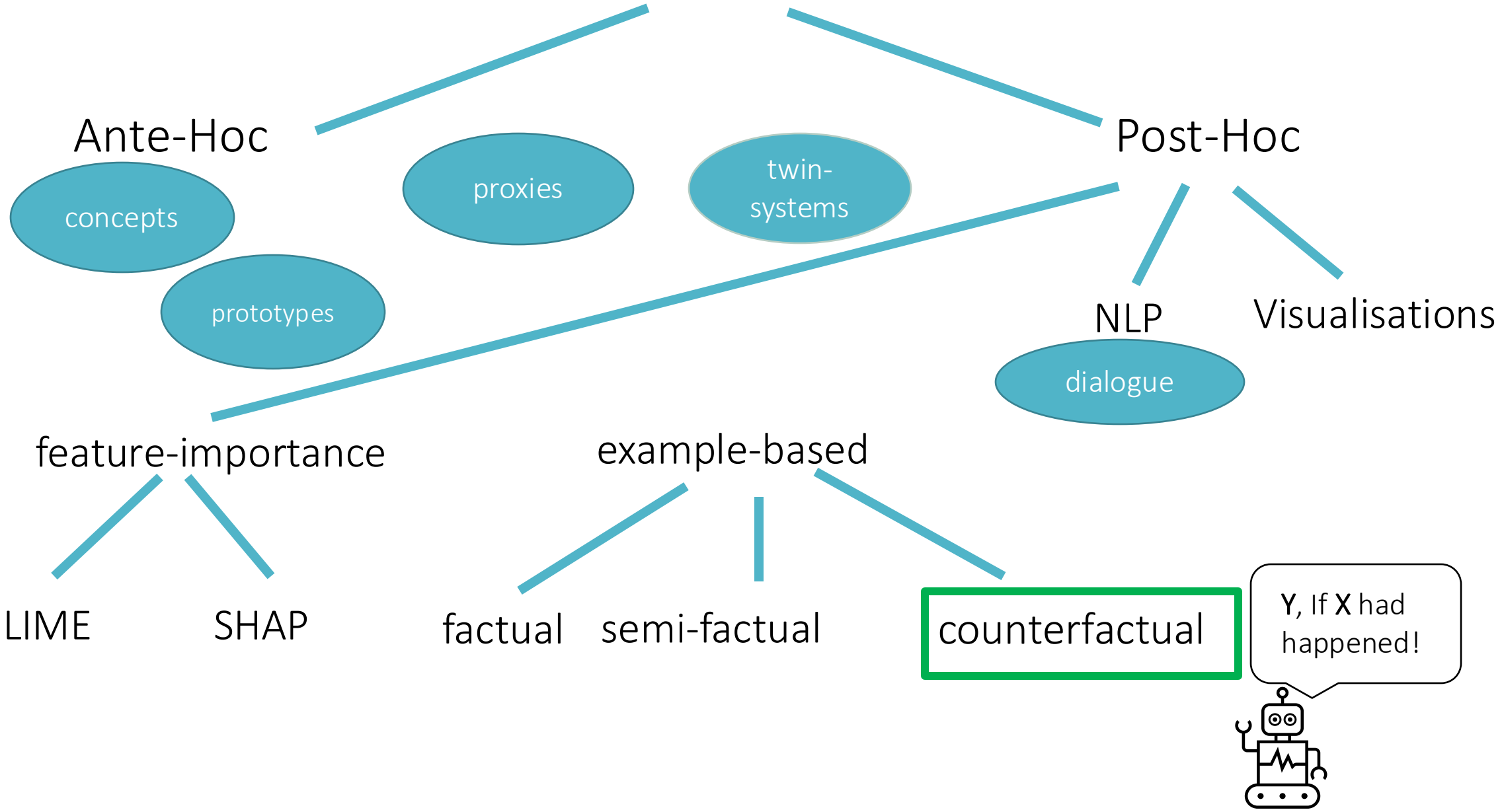
https://github.com/andreArtelt/IJCAI24-CF_Tut
https://github.com/ukuhl/AAAI_CF_Tutorial
https://github.com/andreArtelt/ICPR26_CF_Tutorial



Case-1: *Interpretable Event Diagnosis*

Based on Counterfactual Explanations

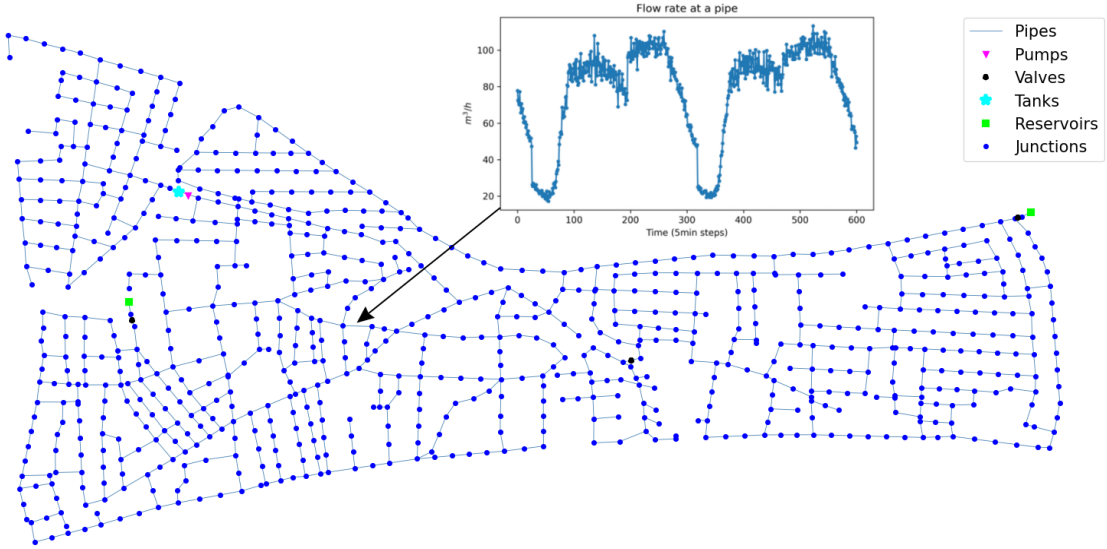
Overview of XAI Methods



Event Diagnosis in Water Distribution Systems

- Operation of complex systems under sparse knowledge
  - Few sensors + lots of uncertainty
- Event Diagnosis = Detection + Analysis of Event
  - Leakage detection & localization
  - Sensor faults
  - Contaminations
  - ….
- Human judgment and experience

=> *Event Diagnosis tools to support human operators*

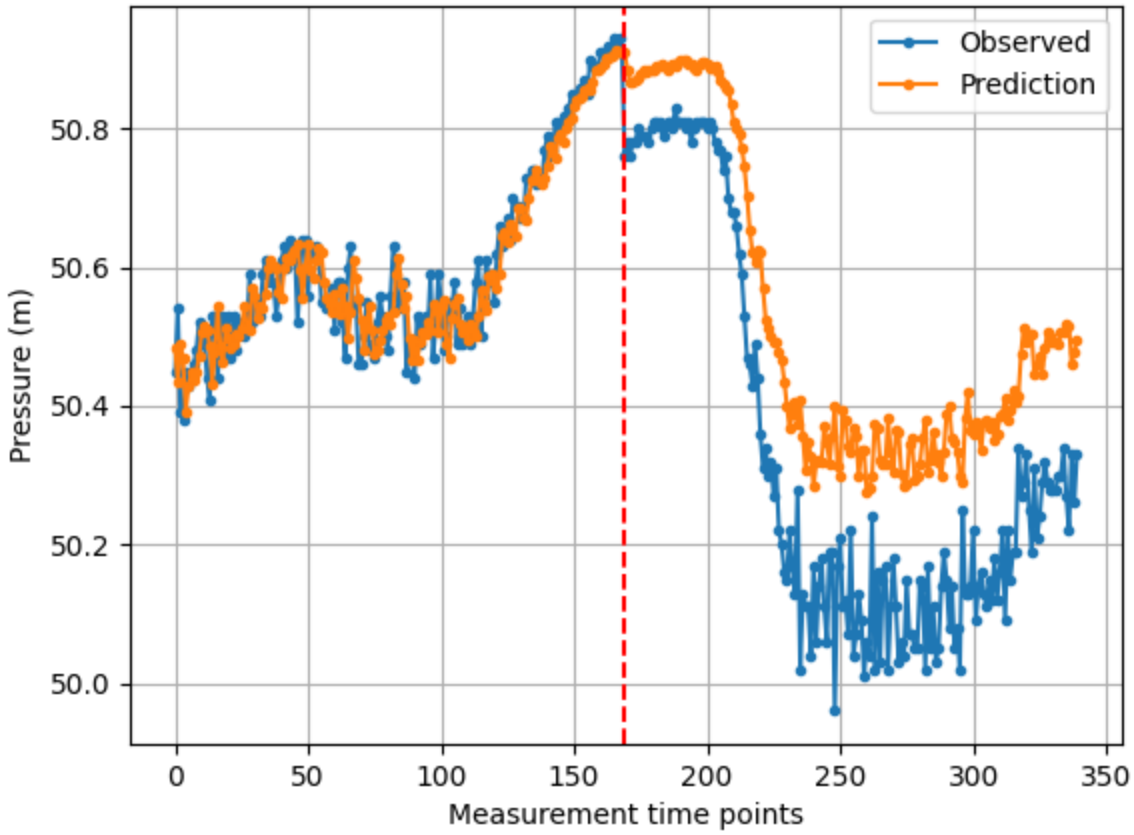


Event Diagnosis Methods

- Classic methods and new AI-based methods
  - Data-driven methods
- *Residual-based methods*:
  - Compare observations to predictions
  - Large deviations => Raise alarm
  - Similar to human intuition

$$\|f(\mathbf{Y}_{[k-T-1,\ldots,k-1]}) - \vec{y}(k)\|_\infty > \theta$$

Simulator, Surrogate model, ….

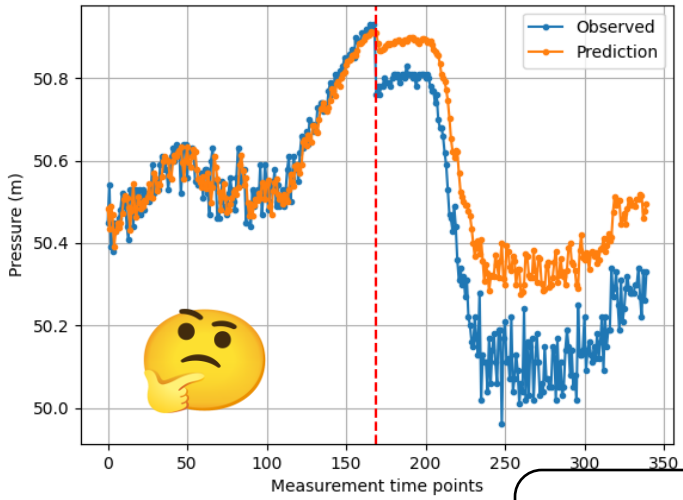


Interpretability of (AI-based) Methods

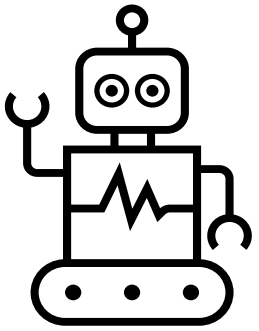
- How does $f\left(\mathbf{Y}_{[k-T-1,\ldots,k-1]}\right)$ work?
- *Lack of transparency => Trust issues*
  - Human operators want to understand the reasoning
- Use XAI to provide insights
  - **Explain the outputs**
  - **Understand their reasoning**

    => Create trust
- What and how to explain

  => *Pick the right XAI method!*



Because of **X**!

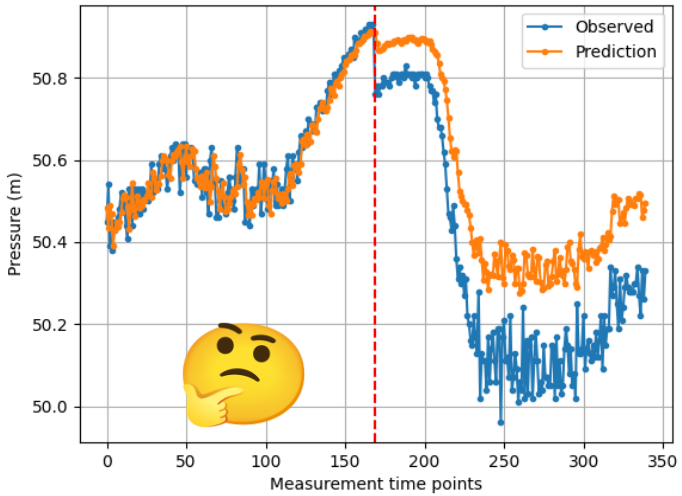


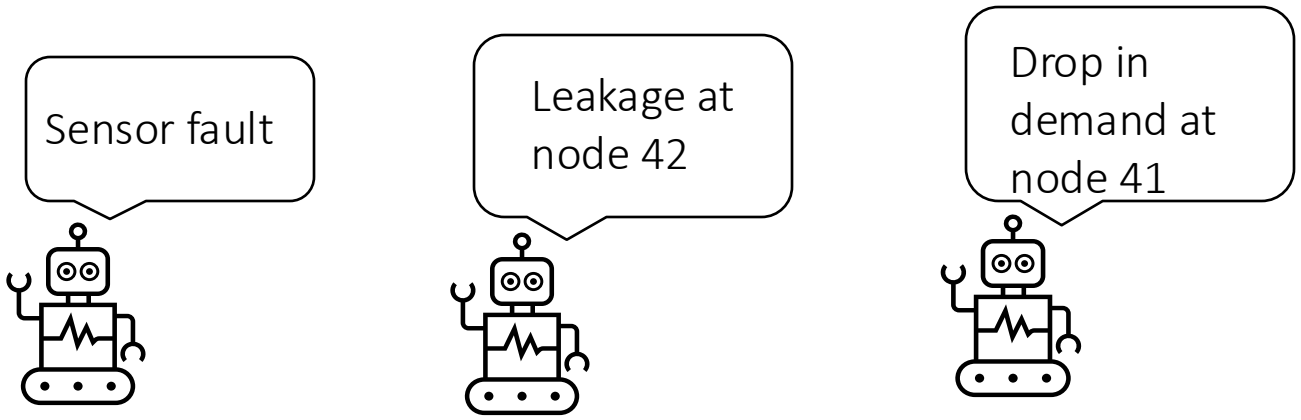
What and How to Explain?

- Contrasting explanations
  - Why prediction differs from observation?
- Highlight feature importances and contributions
  - SHAP

    => Provides a sensitivity analysis (w.r.t. the difference)
- Reasoning over alternative worlds (inputs & assumptions)

  => *Counterfactual Explanations*



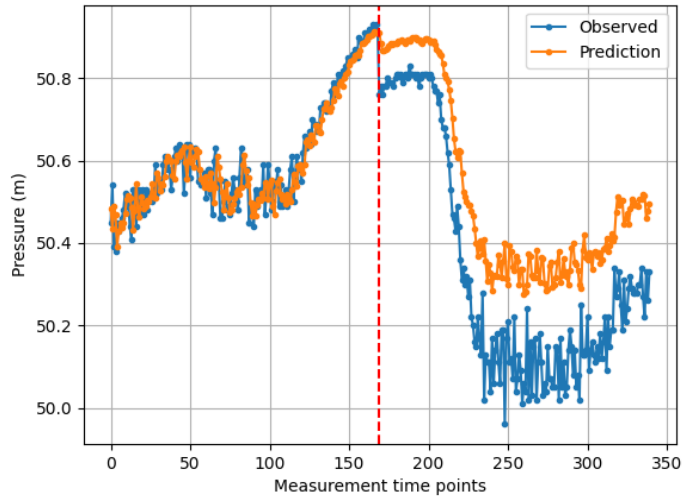


Multiple possible explanations!

Reasoning Over Alternative Worlds

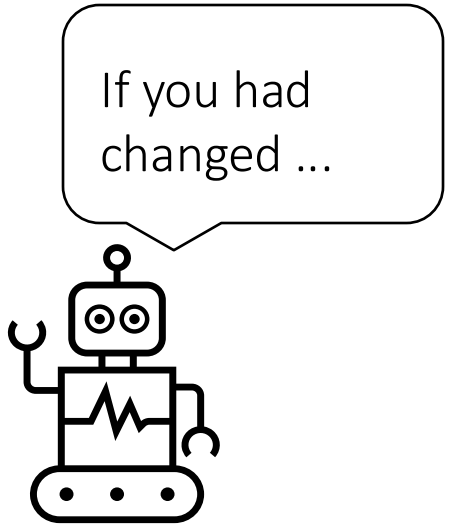
$f(\mathbf{Y}_{[k-T-1,\ldots,k-1]})$ relies on its input + additional assumptions

- Past/Other sensor readings
- Demand estimates
- Parameters of hydraulic laws
- ....



Change prediction of *input/assumptions* $f(\mathbf{Y}_{[k-T-1,\ldots,k-1]})$

=> **Find most "reasonable" changes to "undo" the difference!**



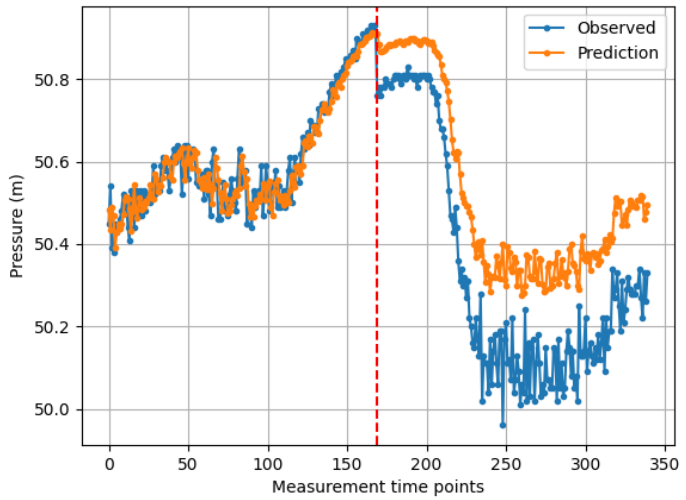
Counterfactual Explanations

More formal:

$$\arg\min_{\mathbf{Y}',\Omega'} ||f(\mathbf{Y}',\Omega') - y(k)|| \leq \theta$$

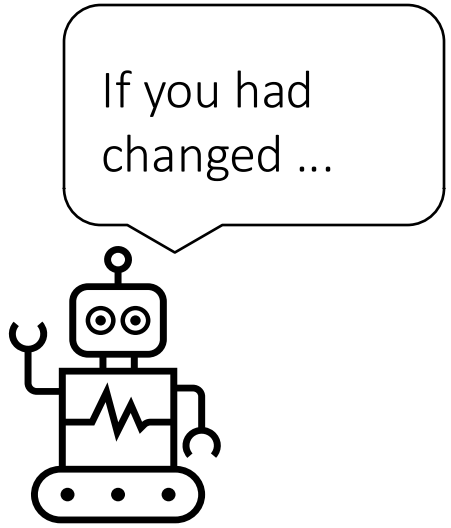
Past/Other sensor readings

Other assumptions and parameters



Our case study:

- Set of virtual sensors:
  =>Predict sensor readings based on past readings from all other sensors
- Leakages and sensor faults
  => Distinguish between those two event types
- Experiments on L-Town (BattLeDIM) and Hanoi (LeakDB)



Artelt, A., Vrachimis, S. G., Eliades, D. G., Kuhl, U., Hammer, B., & Polycarpou, M. M. (2026). *Interpretable Event Diagnosis in Water Distribution Networks.* Intelligent Systems with Applications

How To Counterfactual Explanations

Virtual sensors for event detectior $\|f_i(\vec{x}^i_{t-1}) - (\vec{y_t})_i\|_\infty > \delta$

Ensemble Consistent Explanations:

- A single explanations for a set of models – "fingerprint" of the event

$$\min_{\gamma \in \Gamma(\vec{x})} \theta(\gamma)$$

"Simple" explanation

$$\text{s.t. } \Psi(\gamma, h_i, \vec{x}) \geq 0 \quad \forall\, i$$

Satisfy all moc

$$\min_{\gamma \in \Gamma(\vec{x})} \theta(\gamma) + \lambda \sum_i \xi_i$$

$$\text{s.t. } \Psi(\gamma, h_i, \vec{x}) \geq 0 - \xi_i \quad \forall\, i$$

$$\xi_i \geq 0 \quad \forall\, i$$

Constrair

$$\min_{\vec{x}_{\text{cf}} \in \mathbb{R}^d} \theta(\vec{x}_{\text{cf}}, \vec{x}) + \lambda \sum_i \xi_i$$

$$\text{s.t. } \mathrm{d}\left(f_i(\vec{x}_{\text{cf}}) - y_{\text{cf}_i}\right) \leq \delta_i + \xi_i \quad \forall\, i$$

$$\xi_i \geq 0 \quad \forall\, i$$

Regression metric

Artelt, A., Vrachimis, S., Eliades, D., Polycarpou, M., and Hammer, B. (2022). *One explanation to rule them all – ensemble consistent explanations*. in IJCAI – Workshop on XAI

Computation of Counterfactuals

Virtual sensors as linear regression

- Input: Avg. Sensor readings over time window

$$f_i(\vec{x}^i_{t-1}) = \vec{w}_i^\top \vec{x}^i_{t-1} + b_i \text{ with } \vec{x}^i_{t-1} = \frac{1}{T}\sum_{j=1}^{T}(\vec{x}_{t-j})_{\neq i}$$

Alternatives: Stacked sensor readings, add time, ….

Linear or convex-quadratic optimization problem:

- Efficient solvers exist
- Nearest Unlike Neighbor (NUN) method
  - *Database of previous measurements*
  - *Pick the best solution from this set*

    Ensure plausibility?!

$$\min_{\vec{x}_{\text{cf}} \in \mathcal{D}} (\vec{x}_{\text{cf}} - \vec{x})^\top (\vec{x}_{\text{cf}} - \vec{x})$$

$$\text{s.t. } (\vec{w}_i^\top \vec{x}_{\text{cf}} + b_i - y_i)^2 \leq \delta_i + \xi_i \quad \forall\, i$$

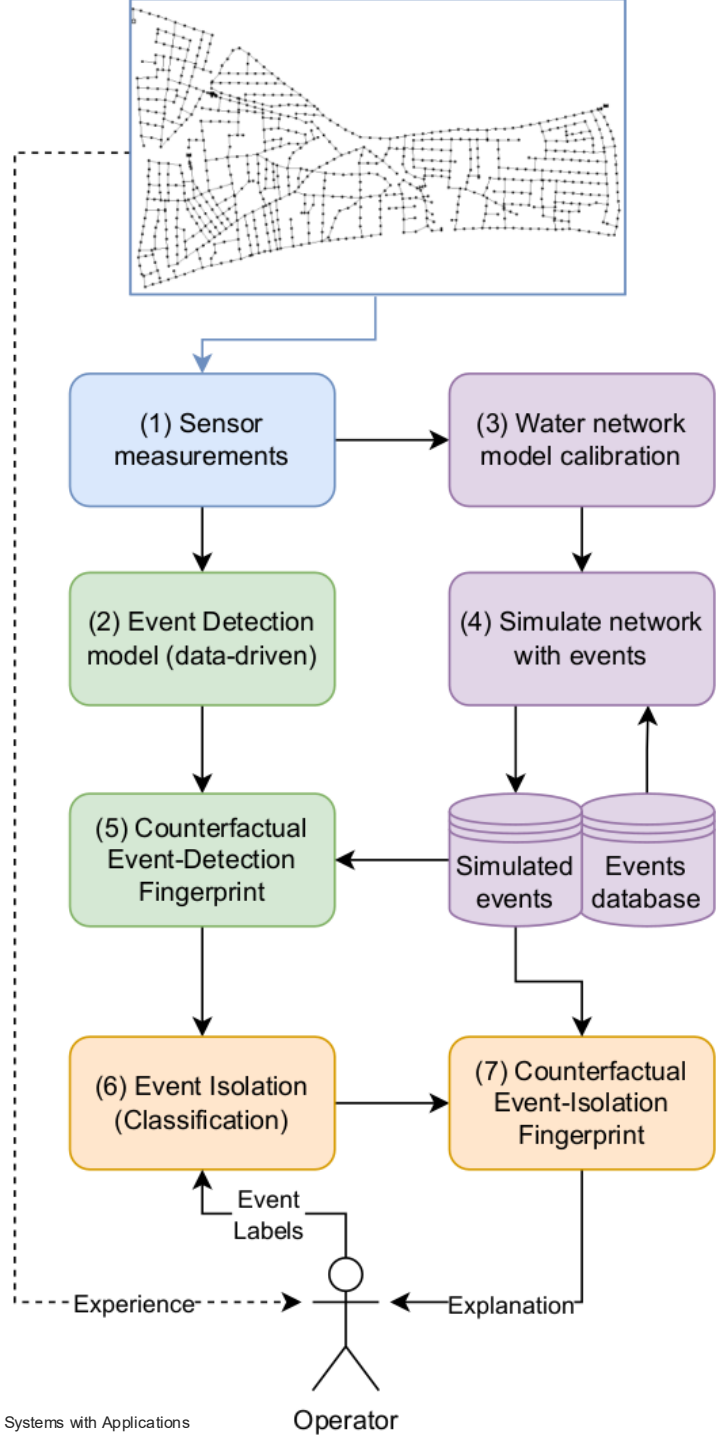
$$\xi_i \geq 0 \quad \forall\, i$$

Event Diagnosis Case Study

- Case study on L-Town variant (only Area A) from BattLeDIM
  - 29 pressure sensors – virtual sensors (linear regression)
  - Different types of sensor faults vs. Leakages
  - *Event Detection + Event Isolation*

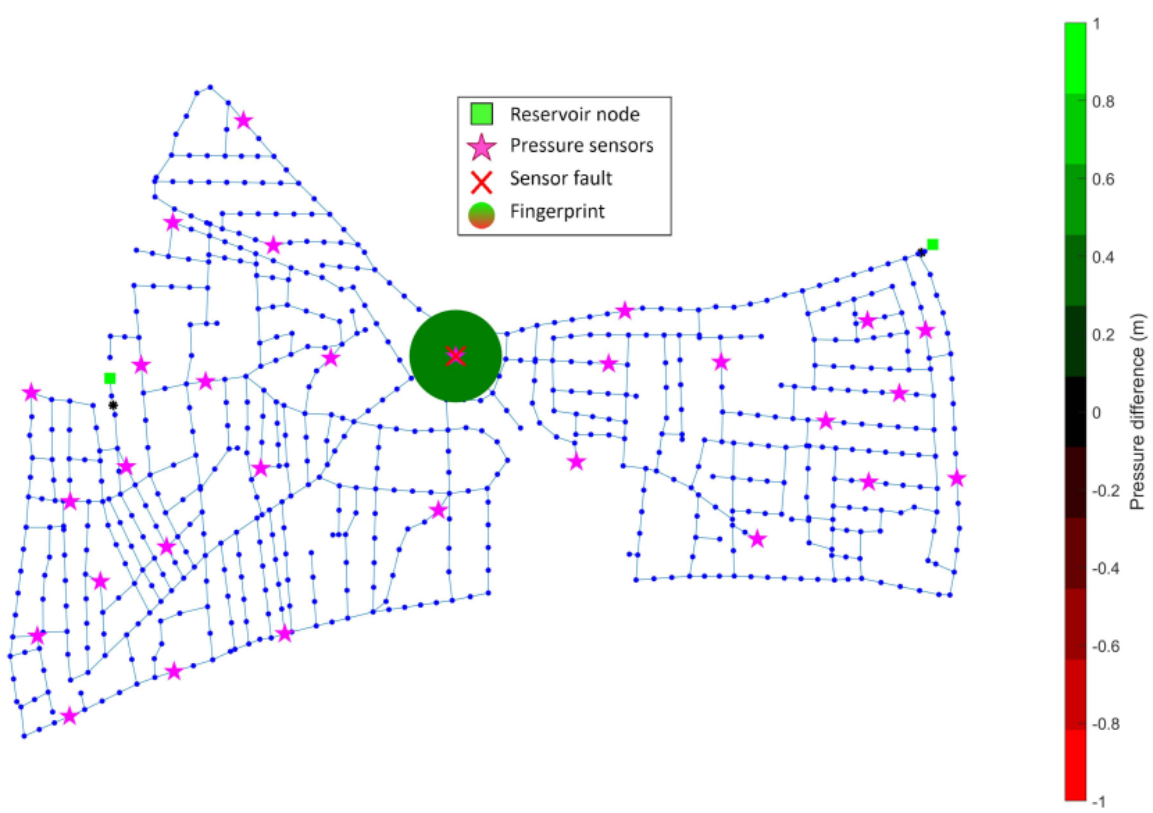
    => Classifier for event: sensor fault vs. Leakage
  - *Counterfactual Fingerprints*

    => Ensemble consistent counterfactuals

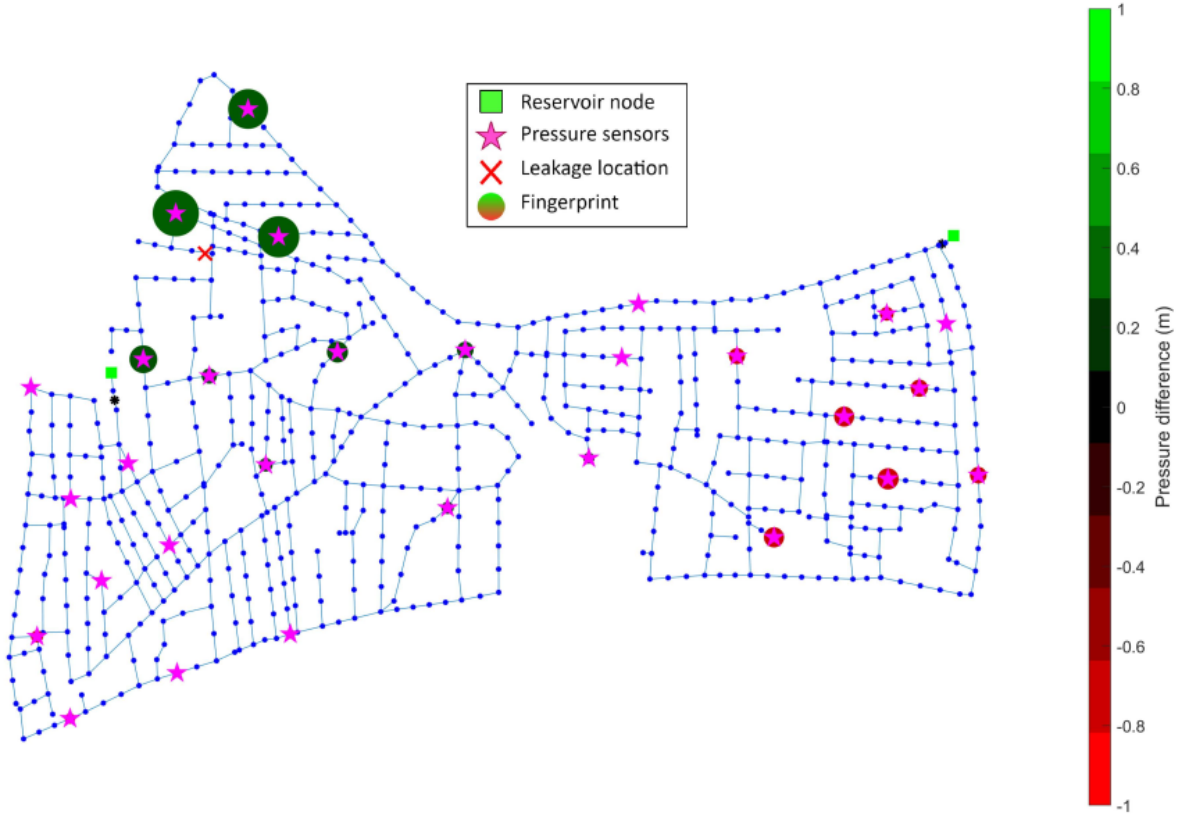


Artelt, A., Vrachimis, S. G., Eliades, D. G., Kuhl, U., Hammer, B., & Polycarpou, M. M. (2026). *Interpretable Event Diagnosis in Water Distribution Networks.* Intelligent Systems with Applications

Visualization of the Counterfactuals



Sensor fault at sensor 'n342'.



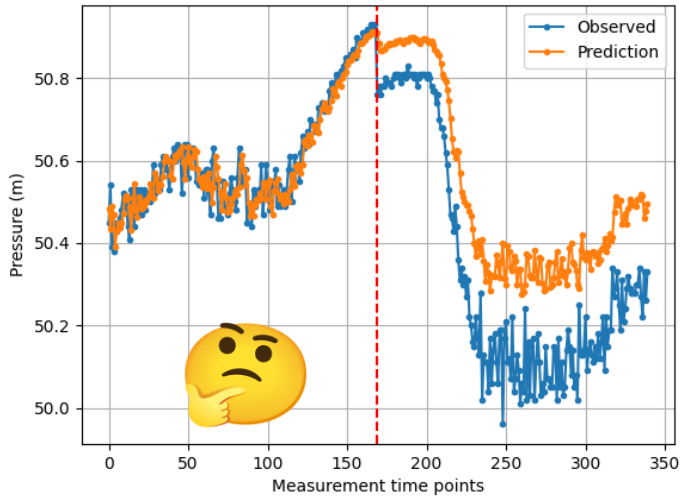
Leak near node 'n66'.

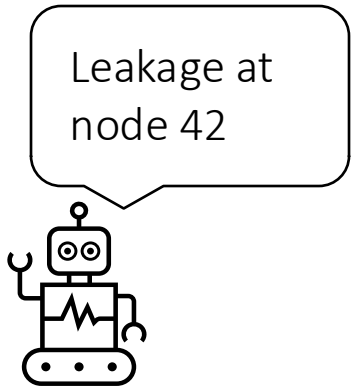
Summary

- XAI for interpretable event diagnosis
- Counterfactuals (XAI) for reasoning over events
- Incorporate more *domain knowledge* and *time* into XAI methods

  => context-aware XAI
- XAI in water quality
  - Different + more complex dynamics
- How to evaluate?
  - How to visualize & communicate?

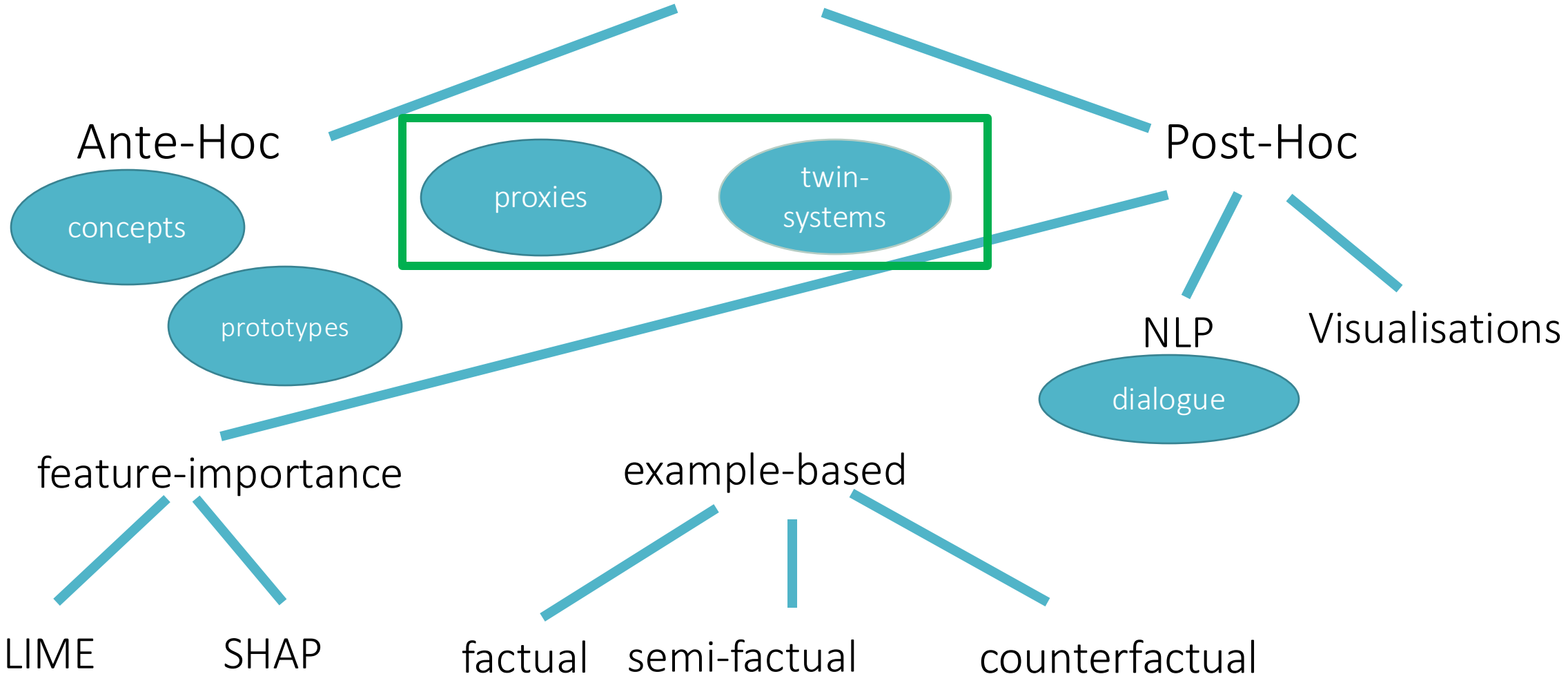
    => We need benchmarks!





Case-2: AI-based *Surrogate Models for What-If Analysis*

Overview of XAI Methods



The Limits of Simulators

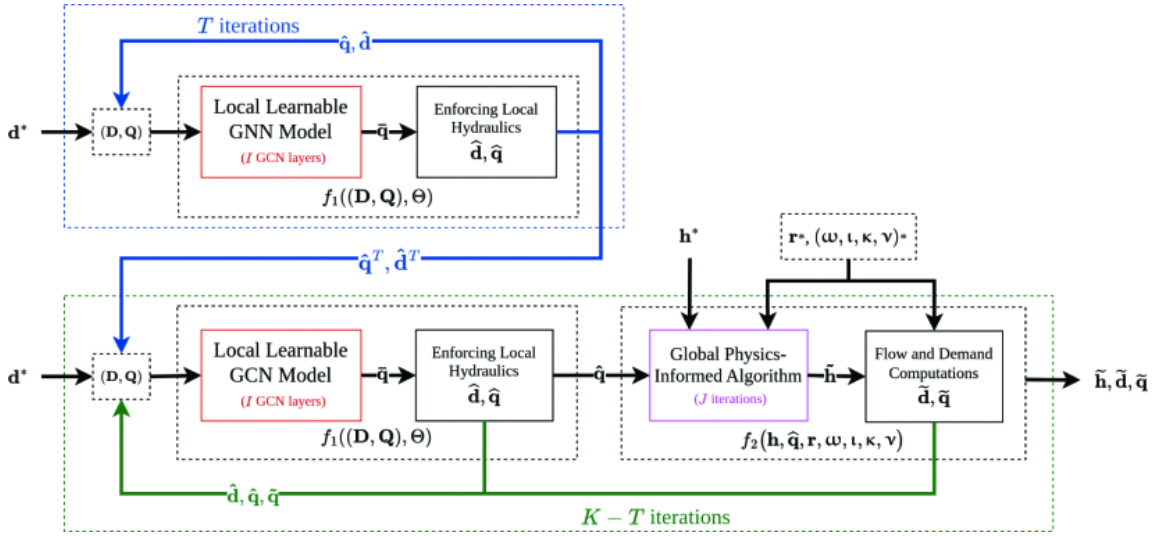
- Many applications *require multiple simulation runs*
  - Sensor placement
  - Event localization
  - Network rehabilitation and planning
  - ...
- Limits of simulators:
  - Computationaly costly
  - Full knowledge assumption
  - ...

Surrogates to the Rescue

- Surrogate model: ML-based approximations of simulations

=> Physics-informed Graph Neural Networks

- Properties:
  - Computationally fast
  - Differentiable wrt. inputs and system parameters
  - (Handler uncertainties)



I. Ashraf, A. Artelt, B.Hammer (2025) Scalable and Robust Physics-Informed Graph Neural Networks for Water Distribution Systems, IEEE IJCNN

Surrogates to the Rescue

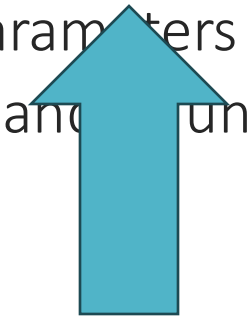
- Surrogate model: ML-based approximations of simulations

=> Physics-informed Graph Neural Networks

- Properties:
  - Computationally fast
  - Differentiable wrt. inputs and system parameters
  - (Handle uncertainties)



- Fast scenario evaluation (e.g., Monte-Carlo)



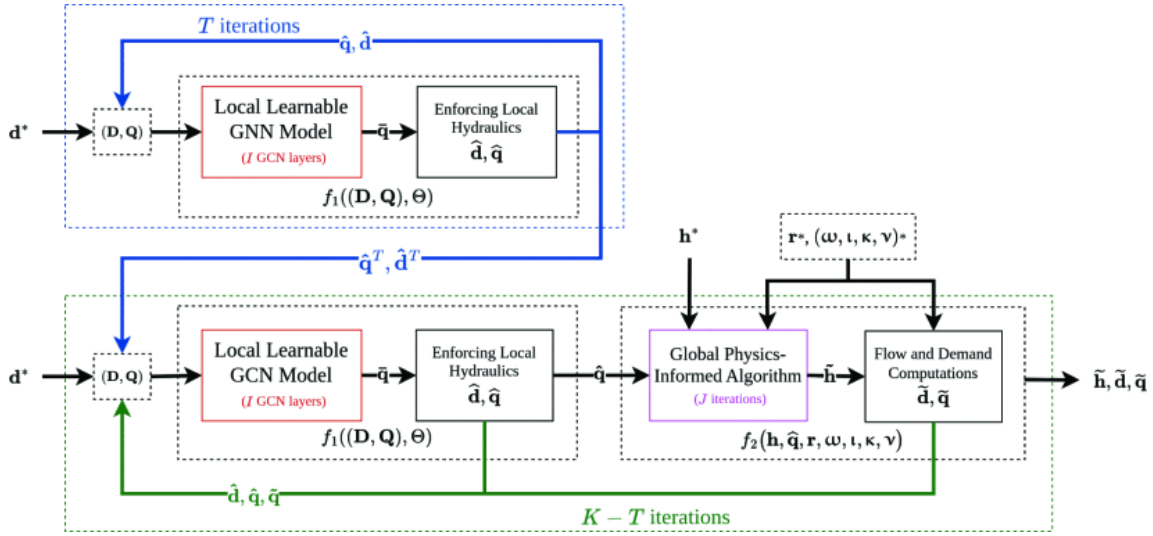
- Sensitivity analysis & Event localization
- Rehabilitation (e.g., finding optimal pipe diameters)

Summary

- Many tasks require efficient What-If analysis

=> *ML-based surrogates can help*

- Physics-informed Graph Neural Networks
  - Fast
  - Differentiable
  - (Uncertainty handling)
- How to build and calibrate surrogates?
  - Data-hungry methods
  - Uncertainties still challenging
  - Benchmarks needed!

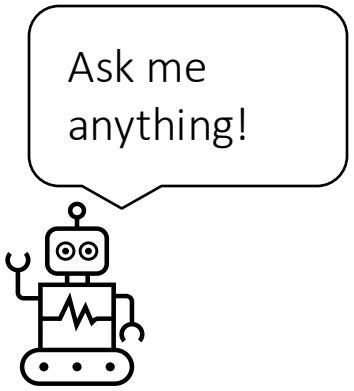


Conclusion & Outlook

- Huge potential of AI/ML in WDS
- Trustworthiness required
  - XAI makes outputs transparent and traceable
- *Counterfactuals* & *Surrogates*
  - Huge potential
- XAI is still a young filed with many open challenges:
  - Integration of system dynamics and physics
  - Uncertainty handling
  - …

- Any questions left?
- Interested in an academic/industrial collaboration?
- Job offer
- ….

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Hands-On in Python

URL: https://github.com/andreArtelt/IAHR26ExplainableML4WDS